



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 30-306-1
19-07-2022

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SPECIFICATION

FOR

PERSONAL PROTECTIVE EQUIPMENT

(HEAD PROTECTION HELMET)

Issue: July-2022/ Rev- 1



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1. SCOPE

This specification covers the minimum technical requirements for design, manufacturing, testing, and supply of protective helmets as personal protective equipment (PPE). The required helmets are intended for protecting the head of workmen of Egyptian Electricity Distribution Companies (EDCs) from falling objects and against the hazard of a fall from a height. Protective helmets are required to reduce electrical shock when workmen near bare conductors and prevent the bare conductors from touching their head.

2. APPLICABLE CODES and STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
ANSI/ISEA Z89.1	Industrial Head Protection
BS EN 397	Industrial Safety Helmets
ISO 3873	Industrial Safety Helmets
OSHA 29 CFR 1910-135	Occupational Safety and Health Standards - Head protection
OSHA 29 CFR 1926-100	Safety and Health Regulations for Construction - Head protection
ASTM F1447-18	Standard Specification for Helmets Used in Recreational Bicycling or Roller Skating



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3. ENVIRONMENTAL CONDITIONS

The safety behavior of the required helmets shall be guaranteed under the environmental conditions given in Table (2). Any differences in the guaranteed performance should be clearly set out in the offer.

Table (2)

Minimum ambient temperature	-5 °C
Maximum ambient temperature	50 °C
Maximum relative humidity	85%
Storage temperature	Dry place
Lifetime	Not less than 5 years

4. CONSTRUCTION REQUIREMENTS

4.1 General

- The helmet shell shall be made of treated hard plastic to protect the head and resist heavy impact without breaking. There shall not any metallic components passing through the shell.
- Helmets should be as light as possible without prejudicing design strength and efficiency. No part of the helmet should have sharp edges and the outer surface of the shell should be smoothly finished.
- The helmet inside shall be fitted with harness including cradle to maintain a safety distance between the liner and the body of the helmet (refer to Figure 1).
- The cradle should be made from textile tapes to provide optimum accommodation of the head shape. This material should have acceptable performance with regard to perspiration and irritation.
- The helmet shall be provided with the chin-strap. It shall be attached to the shell, permanently fitted with a fastening device to adjust and maintain tension. The chin strap shall be made of material not less than 12.7 mm in width.

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Figure (1)

4.2 Type of helmet

- According to ANSI/ISEA Z89.1, helmets are manufactured in two types:

Type I	Helmets are intended to reduce the force of impact resulting from a blow only to the top of the head
Type II	Helmets are intended to reduce the force of impact resulting from a blow to the top of the head or sides of the head (refer to Figure 2).

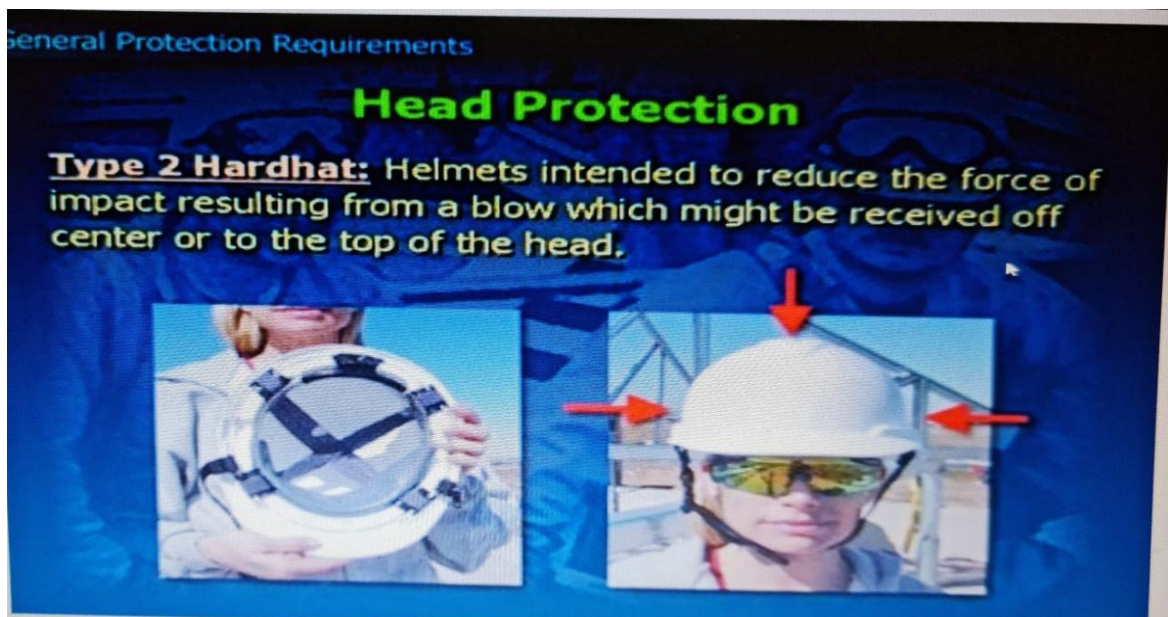


Figure (2)



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- Unless otherwise specified by __ EDC, the required helmet shall be type II. This requirement is fulfilled in EN standard through Lateral Deformation (LD). Type II (or LD) shall be marked on the helmet according to the relevant standard.

4.3 Electrical Class

- According to __ EDC requirement, the electrical class of the helmet can be Class G or Class E as per the following ANSI/ISEA Z89.1 classification:

Class G (General)	Class E (Electrical)
Suitable for low-voltage applications	Suitable for medium-voltage applications
Intended to reduce the danger of contact with low voltage conductors. Test samples shall be proof-tested at 2200 volts (phase to ground).	Intended to reduce the danger of contact with higher voltage conductors. Test samples are proof-tested at 20,000 volts (phase to ground)

5. MARKING

- Each helmet shall be clearly, indelibly and permanently marked by the manufacturer. The nameplate shall include at least the following information:
 - 1) Manufacture's name or trade mark
 - 2) Size or size range
 - 3) Applicable Type and Class
 - 4) Reference standard
 - 5) Serial no.
 - 6) Manufacture and Expire dates



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6. TESTING and INSPECTION

6.1 General

- All tests shall be carried out by and at the expense of the tenderer who shall supply all the apparatus, instruments and equipment.
- Tests shall be carried out by an accredited independent testing laboratory (or other testing laboratories accepted by __ EDC).

6.2 Tests

- Helmets shall be subjected to the following tests in accordance with the latest relevant standard and as specified herein.

Test	Class G	Class E
Electrical Insulation	The helmet shall withstand 2.2 kV (rms) AC, for 1 minute. Leakage shall not exceed 3 mA.	The helmet shall withstand 20 kV (rms) AC, for 3 minutes. Leakage shall not exceed 9 mA.
Shock Resistance	The maximum force transmitted to a rigidly mounted headform on which the helmet is fitted shall not exceed 4450 N. The averaged values shall not exceed 3780 N	
Penetration Resistance	The penetrator shall not make contact with the top of the test headform	
Flammability	No flame shall be visible 5 seconds after removal of the test flame	

7. SUBMITTALS

7.1 Submittals required with tender:

- The supplier shall complete and return one copy of the attached Data Schedule.
- Technical data and original catalogues shall be submitted with the offer.
- **One specimen** of the offered equipment shall be submitted with the offer for evaluation prior to issuance of the purchase order. The specimen should correspond in all respects with the material they intend to supply. Dimensional verification, indelible markings and finishing shall be checked.



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- A valid approval certificate issued by Egyptian Electricity Holding Company.
- Copy of test reports.
- List of deviations & clauses to which exception is taken (if any).

7.2 Submittals required following award of contract:

- Details of manufacturing and testing schedules
- Packing details.
- The supplier should provide all the instructions of use/storing in (Arabic and English).

8. GUARANTEE

- The supplier guarantee the supplied materials against any defects arising out of faulty design or workmanship, or of defective material for a period of one year at least from the date of delivery.

9. PACKING and DELIVERY

- Helmets shall be packaged in an individual container or package of sufficient strength to properly protect the product from damage. The outside of the container or package shall be marked as specified in relevant standard.
- The type of packaging suitable for transport shall be defined by the manufacturer.
- At the request of the customer or according to government specifications any additional or amended instructions shall be included in the package.

10. TECHNICAL DATA SCHEDULE

- The tenderer must fill in thoroughly the attached technical data schedule.
- Any offer does not accompanied with clear and complete technical data schedule shall be rejected.



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TECHNICAL DATA SCHEDULE
PERSONAL PROTECTIVE EQUIPEMENT (PPE)
(Head Protection Helmet)

-- EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	-- EDC Requirement*	Vendor Offer**
3	Manufacturer's name	specify	
4	Helmet Type	Type II	
	Helmet Class	☐ Class G ☐ Class E	
5	Country of Origin	Specify	
6	Applicable Standard	☐ ANSI/ISEA Z.89.1 ☐ BS EN 397	
	Environmental specifications	– Operating temperature – Humidity – Storage temperature	
	Material of shell	Specify	
	Helmet construction	Please refer to clause 5	
	Size range	Specify	
	Universal link	– Material – Length	
	Insulated stick	– Material – Length	
	Physical specification	– Length – Diameter – Weight	



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10	Year and Month/Quarter of Manufacture	Manufactured within two years before the bid closing date.	
11	Marking on the equipment	Please refer to clause 6	
12	Test Reports	Please refer to clause 7	
13	Specimen	Please refer to clause 8	
14	Guarantee	Please refer to clause 9	
15	Packing and delivery	Please refer to clause 10	
16	Submittals	Please refer to clause 8	

(*) - Values to be provided / proposed by the vendor.

(**) – Please, provide explanations for deviations if any. (Use separate sheet if necessary.)

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		